FORMATIVE ASSIGNMENT

[Eskuul]

- **Description of the concept of their Mission;**

  I aim to solve the unequal access to quality education in Somaliland by creating a digital learning platform. The platform will offer a structured guidance, study materials and self assessment tools to help students make the most out of their high school journey with high hopes to deploy the platform by the end of December, and have 500-2000 students registered by the end of the academic year.

  This mission is applicable in the education sector in Africa due to the problem of unequal access to resources and qualified teachers. Lack of qualified teachers and modern technology that helps students even after school hours are limited in many countries in the region. By centering on Somaliland, the project helps address these issues locally, while also playing a role in reducing inequity in education throughout Africa.

- **Problem statement:**

  The high school students of Somaliland are in a difficult situation to prepare for the national exams. They lack access to trained teachers, up-to-date textbooks and digital materials in a uniform manner. National examinations ask for knowledge from several years of teaching the subject, but most students are not provided systematic guidance on how to prepare for them. This causes poor performance, stress, and lack of access to higher education.

  Who: High school students are the main beneficiaries in Somaliland, specifically those in rural or low-income communities. Teachers and families are also affected, as exam results determine people's future education and employment opportunities.

  What: The problem is less equal access to the quality of teaching and learning materials, which means that students are not well prepared for

national examinations.

When: The problem occurs during the last years of secondary school when students have to prepare for Somaliland National Examination, which is a criterion for admission to higher education.

Where: The issue takes place throughout Somaliland, with the rural and low-resourced areas being the most affected.

Why: The relative immaturity of study programs as well as limited capacity of teaching staff contribute to learning gaps which accumulate over time. The gaps that exist when the students reach the examination stage are too great, thus restricting their chances of acquiring higher education qualifications and future employment opportunities.

Software Development Model

Agile development model is the most applicable in this project. Agile enables gradually release which can be tested directly with students and teachers. This is noteworthy since the real-user feedback in Somaliland will inform the changes in content and features.

Agile steps that will be applicable will include:

- **Requirement Gathering**: Preliminary discussion of curriculum needs with teachers and students to get them on the same page of the most urgent requirements.
- **Planning :** The project is broken into small and time bound sprints like PDF upload module, quiz engine, and student tracking.
- **Design and Development:** Development of working prototypes of every feature.
- **Testing:** Implementing modules to small groups of students to test their functionality and precision.
- **Review and Feedback:** A collection of feedback at the end of every sprint with corrections.
- **Deployment:** Implementation of the complete web app in the schools and continued updating.

The flexibility of Agile is useful in the education setting where students have different needs and technology constraints are actual.

Hypothesis of the Solution

The hypothesis will be that high school students in Somaliland will be better prepared to take exams once they are given structured summaries and

quizzes in a single platform. This expectation can be proven using evidence elsewhere. According to UNESCO (2023), students with organised learning tools had on average 22 percent higher scores. The Education Endowment Foundation (2020) conducted a meta-analysis that showed that a consistent self-quiz could enhance memorization by up to 18 percent.

It is expected that the study will have the expected level of improvement in the scores of students on mock exams using the platform in one academic term. This will be expected to decrease the performance gap between urban and rural schools in the long term due to increased accessibility to consistent study resources.

References

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Software Requirements Specification (SRS)

Title: Eskuul

Prepared by Hamse Hassan Jama

Organization : ALU (student project)

date created: 09/25/25

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Revision History

Name	Date	Reason For Changes	Version
Hamse	25/9/2025	Initial draft	1.0
Hamse	17/10/2025	UML Diagrams	2.0

1. Introduction

1.1 Purpose

This software requirements specification document (SRS) is intended to outline the project of digital learning platform. It is designed to reduce the education inequality in Somaliland highschools. The primary goal is to ensure the availability of the same high quality educational content and resources to students across the country.

1.2 Document Conventions

The document follows simple consistent formatting. It uses standard font size, black text, and Montserrat font for all text. This SRS document will use a number based

1.3 Intended Audience and Reading Suggestions

This SRS serves several stakeholders involved in developing and delivering the project

1. **Developers:** Refer to the “System Features” and “External Interfaces" more in depth understanding.
2. **Project Managers:** Start with the "Overall Description" to get the project context, scope and constraints.

3. **Marketing Team:** Start with the "Introduction" section to understand the project's goals then refer to the "System Features" for system functionalities.
4. **Users and Testers:** Focus on Authentication (register/login), dashboards, CMS

1.4 Product Scope

The product is a web application that is responsive in the use of modern browsers. It offers PDF summaries and quizzes in the style of similar tools as Cambridge in 5 Minutes and FreeExamAcademy. The system will be customized according to Somaliland curriculum and national examination standards.

2. Overall Description

2.1 Product Perspective

It is a web-based application where students can use it independently. This product is a standalone product and is not part of an existing system. It augments the current classroom experiences but does not substitute teaching.

2.2 Product Functions

The platform will provide the following main functions that would assist students and teachers in Somaliland:

- **Access to PDF Summaries**
    - The curriculum syllabuses by subject and grade as PDFs can be viewed and downloaded by the students.
    - The content will be easily navigated by being searchable and filterable.
    - Notes will be optimized by accessing them at low bandwidth.
    -
- **Quizzes and Auto-Grading**
    - Every PDF summary will be associated with a series of multiple choice questions.
    - The auto-gradation of quizzes will be realized in real-time and will offer immediate feedback and correct answers.
    - Whenever possible, students will be explained why they have wrong answers.

- **Student Progress Tracking**

    - The system will capture the attempts to take quizzes, scores and completion.
    - Individualized student progress report will be displayed to the students.
    - Reports will be used to point out areas that will be used to study.

2. **Responsive Study Interface**

    - The platform will have a straightforward interface that it can use when studying long without distraction.
    - The content will go automatically to various screen sizes, and this also includes inexpensive smart phones.
    - Usability characteristics like fonts of different sizes will be accommodated.

2.3 User Classes and Characteristics

- **Students (Primary Users)**

    - The system will be used to give access to PDF summaries and quizzes to students.
    - Entry-level computer literacy will be available to most users hence interfaces will be straightforward and user-friendly.
    - Most of them will be stuck on smartphones and low internet connectivity and hence mobile optimization is essential.
    -
- **Teachers (Content Creators)**

    - Instructors will prepare and post some PDF summaries and quizzes.
    - They will be digitally mediocre and require clear guidelines on how the content will be submitted.
    - Educators can also check analytics as they strive to monitor the way students interact with their assignments.
- **Administrators (System Managers)**

    - Admins will monitor the running of platforms.
    - The responsibility involves user accounts approvals, content moderation and maintenance of systems.
    - Admins will require complete access to usage data, security logs and error reports.

2.4 Operating Environment

1. **Platform**

    - The solution will come in a form of a responsive web application that can be accessed using the modern devices.
    - Lowest supported resolution: 360x640 pixels (low-end smartphones).

2. **Browsers Supported**

    - Chrome (latest two versions)
    - Firefox (latest two versions)
    - Safari (latest two versions)
    - Microsoft Edge (latest two versions)

3. **Backend Environment**

    - Server: Node.js with Express.js for REST APIs.
    - Database: PostgreSQL for structured storage of user profiles, quizzes, and performance data.
    - File Storage: Cloud object storage (e.g., AWS S3) for PDFs.

4. **Hosting and Infrastructure**

    - Cloud-based deployment (e.g., AWS, Azure, or DigitalOcean).
    - Load balancing and auto-scaling for peak exam seasons.
    - Daily backups for reliability and disaster recovery.

2.5 Design and Implementation Constraints

- **Connectivity Limitations**

    - A large portion of the users will have low or poor internet connection to the system. The system should be able to trim down file sizes, put most of the in-use resources on a cache, and work with unconnected work at intervals.
- **Device Constraints**

    - It should be able to work efficiently with cheap smartphones that have low RAM and storage. There will not be heavy reliance on animation or huge media file.

- **Content Constraints**
    - The first content will be in the form of PDF summaries and quizzes only.
    - Teachers or students are not allowed to post their projects open to everyone to ensure it is of quality and it is directed towards the national curriculum.
- **Security and Privacy**
    - The data of the user must be kept in a secure location according to the international best practices.
    - It must be secured with authentication so that it cannot be lost to unauthorized users but be very easy to access by the entry-level users.
- **Scalability**
    - The system must be in such a way that it can grow in users and also the content volume especially during exam seasons.
- **Regulatory Constraints**
    - The content should not contradict with the national curriculum of Somaliland.
    - Hosting has to be secure according to the local data handling laws.

2.6 User Documentation

The site will contain readable and realistic user documentation to assist all categories of users:

- **Online User Guide**
    - They will be taught in a step-by-step manner diving into how the students can avail PDF summation, passing quizzes, and seeing progress reports.
    - The teachers will be provided with a report on the process of uploading PDF summaries with creation of quizzes and student performance statistics.
    - A different guide will be used with administrators that will entail the management of accounts, content moderation, and system monitoring.
- **Help and Support Section**

    - A help page will be created to answer any question that can be commonly asked or frequently asked (FAQ).
    - The issues that will be covered are account registration, troubleshooting quiz submission, and recovery of a lost password.
    - Short tutorial videos will be provided through links to provide assistance to entry-level users.
    - Tooltips and In-App Guidance
    - The initial user encounter with the platform will be simplified with help of on-screen tooltips.
    - Sample: questions on regarding how to begin a quiz or where to get notes.

- **Distribution of Documentation.**

    - Any user documentations will be stored immediately in the platform and made phone friendly.
    - The user guides will be in PDF formats that have the option of downloading to be used offline.

2.7 Assumptions and Dependencies

There are a number of assumptions and external factors on the success of this project:

- **The Availability of the Internet and Devices.**

    - The assumption is that the students will have the simple access to the smartphones or computers and the internet connection.
    - Although this connectivity can be intermittent, the system must have connection at least on a periodic basis through online connections.

- **Curriculum Stability**

    - The platform presupposes that the national curriculum will stay rather unchanged in the course of the development and the first deployment years in Somaliland by the Ministry of Education.
    - The significant changes in the curriculum would demand the alterations to PDFs and quizzes.

- **Teacher Involvement**

    - The project will rely on qualified teachers who will offer correct PDF summary and quizzes.
    - In the absence of teacher involvement, the development of the content will be reduced, making the platform ineffective.

- **Third-Party Services**
    - At least AWS, Azure, or DigitalOcean will serve as the outsourced providers of hosting services, cloud storage, and database infrastructure.
    - Any interruption of service or downtimes by these providers may impact on availability

.

- **Adoption by Students**

    - The assumption is that students will be encouraged to use the platform as an addition into their education.
    - The solution would have low adoption rates resulting in the minimization of its effects.

3. External Interface Requirements

3.1 User Interfaces

The site will be a web application, which is responsive and can be viewed through the use of modern devices using browsers on the desktop and mobile platform. The interfaces will focus on simplicity, ease of access, and usability by the students and teachers.

Student Dashboard

- Home screen containing access to full access PDFs of available summaries, subject and grade level sorted.
- Question and answer section; this is where students undergo multiple choice questions with immediate feedback.
- Tracker of progress with quiz done, marks, and areas of improvement.
- Basic information and the performance history has been shown on the profile page.

Teacher Dashboard

- Curriculum compatibility PDF summary upload tool.
- Creator of quiz applications including multiple-choice test items, answer code and subject tagging.
- Students menu analytics display which shows student interactions with uploaded resources.
- Notification section to announce updates or errors that have been flagged.

Administrator Dashboard

- Approving teacher accounts user management interface.
- use PDF summaries and quizzes that have to be reviewed and approved by content moderation tools that follow before publication.
- System metrics (number of active users, most popular subjects, errors).
- The availability of the backup and the system maintenance functions.
- The design will be based on accessibility guidelines of WCAG 2.1 to accommodate the visually impaired students and ensure international best practices. The interface will be based on easy to read typography, regular navigation and mobile first design.

3.2 Hardware Interfaces

The installations and cloud server: the platform will be deployed on a cloud server. They will need only a device which has the access to the internet and has a modern web browser. The system will be supporting low cost smart phones (Android) as well as typical desktops/laptops. The user end will not require any special hardware.

3.3 Software Interfaces

- Browser Compatibility: The system will be compatible with the modern versions of Chrome, Firefox, Edge and Safari.
- Database: User profiles, quiz data and content uploaded by teachers will be stored in Database post gre bleu.
- Server-Side Environment: Node.js and Express.js, both used as backend APIs.
- Authentication: OAuth 2.0 or JWT-based authentication.
- File Handling: PDFs will be hosted and provided on a cloud application object store provider like AWS S3.

3.4 Communications Interfaces

- **Internet Access:** The application needs a good internet connection.
- **Protocols:**
    - Secure client server communication based on HTTPS.
    - Entrap of frontend and backend interaction through REST API.
    - Email communication via SMTP in order to send approval by teachers emails and notifications regarding the system.

4. Requirement Specification

4.0 Functional Requirements

Req ID	Requirements	Description
FR1	**Upload and Manage PDF Summaries**	Admins must be able to upload, edit, and delete curriculum-aligned PDF study summaries.
FR2	Quiz Creation and Grading	Admins must be able to create quizzes linked to topics. The system must auto-grade quizzes and provide instant feedback.
FR3	Student registration and login	Students must register and log in securely to access resources.
FR4	Responsive Interface	Students should be able to search notes by subject and topic and navigate easily.
FR5	Teacher Content Review	A review process should exist to ensure uploaded PDFs and quizzes are high quality and curriculum-aligned.
FR6	Notifications	Students should receive notifications on new content or quizzes.
FR7	Peer-to-Peer Discussion	Students could have access to a moderated forum or discussion board for collaborative learning.
FR8	Analytics Dashboard	Admins should have a dashboard to track system usage, content engagement, and performance data.
FR9	Offline Access	Students could download PDFs for offline use.
FR10	Progress Tracking	The system must track quiz scores and display performance over time to students.
FR11	Search and Navigation	Students should be able to search notes by subject and topic and navigate easily.

4.1 Content Management

- The system will enable the admins to add, update, or delete curriculum-based PDF summaries.
- The system will facilitate metadata-tagging of the objects and topics to enhance the search.
- The system would contain uploaded content safely on cloud storage.

4.2 Quiz and Assessment

- It will enable the admins to make quizzes, which have multiple choices in them.
- The system will be made to mark the quizzes automatically and show the results instantly.
- The system will also explain correct and incorrect answers where necessary.

4.3 User Management

- The system will enable students to be registered and log in safely.
- The system will enable the students to change their passwords.
- The system will be limited to the role-based access of the admins to the features of uploading and editing.

4.4 Progress Tracking

- The system will record scores in the quizzes of every student.
- The system will produce easy progress reports of strengths and weaknesses.
- The system will be enabled to enable the students see their performance history.

4.5 User Interface

- The system will have a responsive mobile and desktop browsers interface.
- The interface will be able to support Somali and English text.
- The interface will provide a distraction free study space.

5. Other Nonfunctional Requirements

Requirement Type	Req ID	Description
Security	NFR1	The platform must authenticate users and protect personal data with encryption.
Performance	NFR2	The system must handle at least 500 concurrent users without performance issues.
Usability	NFR3	The interface should be simple and require minimal training for students with basic digital literacy.
Availability	NFR4	The system should maintain 99.5% uptime on cloud infrastructure.
Scalability	NFR5	The system should support future expansion to 5,000+ users.
Compatibility	NFR6	The system must support Chrome, Firefox, Safari, and Edge.
Auditability	NFR7	Admin activity logs should be available for accountability.

5.1 Performance Requirements

- The system should be able to cater to a population of 500 simultaneous users and have response times of less than 2 seconds.
- The cloud hosting solution has to be in a position to auto-scale to address peak needs.

5.2 Safety Requirements

- The system will do backups of all user data and content on a daily basis.
- The system will have recovery mechanisms to regain service when there is a failure of the servers.
- All files uploaded to the system will be verified to eliminate malicious codes.

5.3 Security Requirements

- The system should have user authentication with secure passwords.
- Any sensitive information will be coded in transit and rest.
- All accountability will see all the activities of the administration logged in the system.
- Only authorized users will be in a position to control study materials.

5.4 Software Quality Attributes.

- As a platform, 99.5% uptime will be maintained yearly.
- The platform will offer a user-friendly interface, which is appropriate among students with low computer literacy level.
- The system will work with the latest versions of the browsers such as Chrome, Firefox, Safari, and Edge.
- This system will be scalable and will offer 5000+ users onboard as more people adopt them.
- The platform will be made in line with accessibility requirements in order to accommodate visually challenged learners.

5.5 Business Rules

- **Role-Based Access**
    - The Admins have the privilege to add, update and remove PDF summaries, quizzes, or data within the system.
    - The students are confined to reading, downloading, and trying quizzes.
- **Quiz Attempts**
    - Learners are allowed to complete each quiz a specified number of times (default is 3) to motivate students to study.
    - A minimum separating time between retakes should be 24 hours.
- **Curriculum Alignment**
    - All the uploaded material should be in accordance with the official syllabus of the Ministry of Education.
    - All off-curriculum content has to be marked as Supplementary so as to prevent confusion.
- **Progress Reports**
    - Only the student can see his or her report.
    - Admins can view aggregated reports anonymously to use in analytics, but cannot view the individual performance of students without permission.
- **Data Retention**
    - The Student accounts that are not used within a period of 12 months shall be archived.

    - The backups should be saved within 90 days.

- **Language Policy**

    - All the official study material should be provided in English and Somali where it is possible.
    - Default language used in Interface is English, but language can be changed depending on the user.

6. Appendix

- Appendix A: Glossary

    - Admin: A user who is authorized to upload, maintain and administer study material and quizzes.

    - Student: A student who is focused on the platform to study the materials, complete quizzes, and track progress.

    - Pdf Summaries: Digital study notes in accordance with curriculum topics.

    - Letter of Progress: Visual summary of work done by the student including quizzes answers and mastery of subjects.

    - Responsive Design: It is a web design that guarantees that the site is usable on both mobile short and tablet devices and on a desktop.

    - Role-Based Access Control (RBAC): This is the method of access control of system depending on user roles.

- Appendix B: UML Diagrams

B.1. Class Diagram

Description:
This class diagram models the structural backbone of the EskuuL platform. It defines the system's key entities, their attributes, operations (methods), and the relationships between them. The design supports core system features such as user authentication, content uploading, quiz management, and student progress tracking.

Main Classes and Their Responsibilities

1. User

- Attributes:
  ID, Name, Email, Password
- Methods:
  signUp(), login()
- Description:
  The abstract parent class for all users in the system. It provides common authentication features and user details shared by Students, Teachers, and Admins.

2. Student (inherits from User)

- Methods:
  viewNotes(), attemptQuiz(), viewProgress()
- Description:
  Represents learners who access educational materials and quizzes. Students can download notes, attempt quizzes, and view their performance reports.

3. Teacher (inherits from User)

- Methods:
  uploadNotes(), manageQuiz(), viewAnalytics()
- Description:
  Represents instructors who contribute content to the platform. They can upload PDF summaries, create or manage quizzes, and review performance analytics of their students.

4. Admin (inherits from User)

- Methods:
  manageAccounts(), approveContent(), viewAnalytics()
- Description:

System administrator responsible for approving uploaded content, managing user accounts, and monitoring system analytics. Ensures that the content aligns with educational standards.
5. PDF Summary

5. PDF Summery

- Attributes:
  fileID, subject, topic, filePath
- Methods:
  download(), view()
- Description:
  Represents uploaded study materials in PDF format. Each file is associated with a teacher and can be downloaded or viewed by students.
- Relationship:
  - A Teacher can upload many PDF Summaries (1..*).
  - Each PDF Summary is uploaded by exactly one Teacher (1).

6. Quiz

- Attributes:
  quizID, title, totalMarks

- Methods:
  gradeQuiz(), getResults()
- Description:
  Represents a test created by teachers to assess student learning. Each quiz can have multiple questions and is attempted by multiple students.
- Relationship:
  - A Teacher manages multiple Quizzes (1..*).
  - A Quiz contains multiple QuizQuestions (aggregation, represented by a hollow diamond).
  - A Quiz can have multiple QuizAttempts linked to students (1..*).

7. QuizQuestion

- Attributes:
  questionID, questionText, options, correctAnswer
- Description:
  Represents an individual question in a quiz. Each question belongs to a single quiz.

8. QuizAttempt

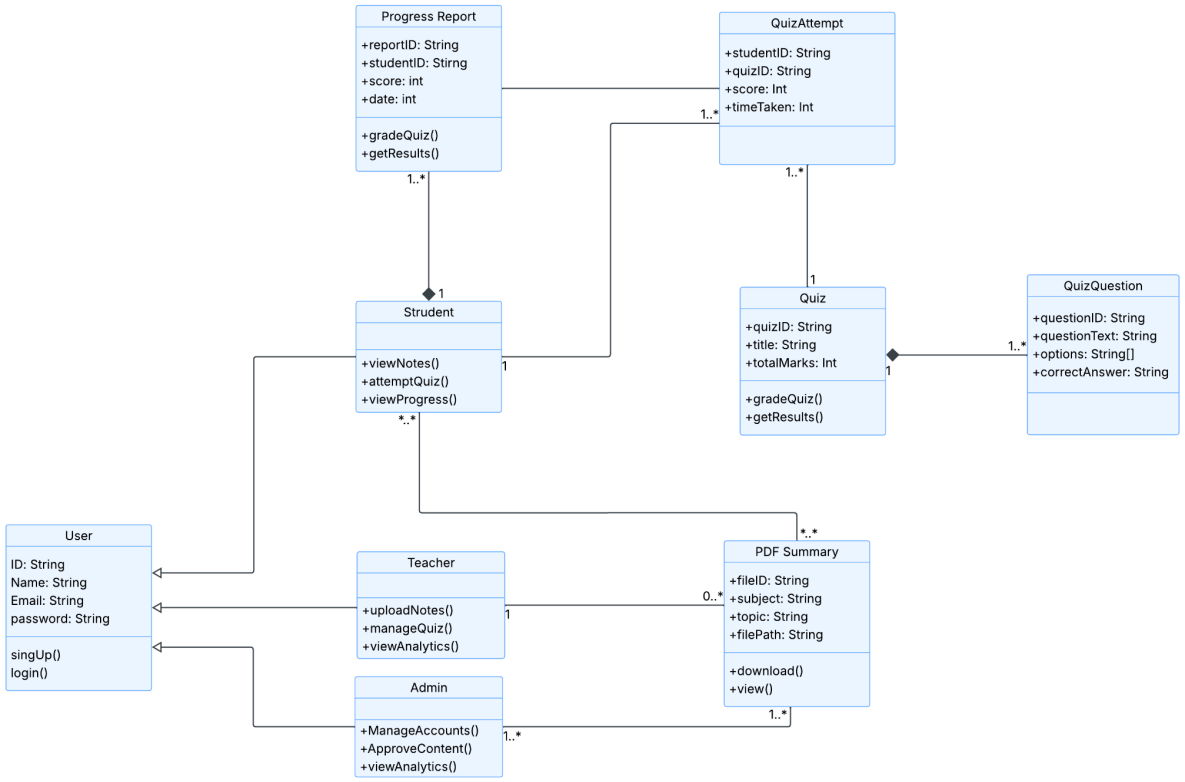
- Attributes:

studentID, quizID, score, timeTaken

- Description:
  Records each student's attempt at a specific quiz, including score and completion time.
- Relationship:
  - Each **QuizAttempt** is linked to one **Student** and one **Quiz** (many-to-one relationship).

9. Progress Report

- Attributes:
  reportID, studentID, score, date
- Methods:
  gradeQuiz(), getResults()
- Description:
  Represents the summarized performance data for a student across various quizzes and subjects.
- Relationship:
  - Each **Progress Report** is associated with a **Student** (many-to-one).
  - It helps track progress over time based on quiz results.



B.2. Use Case Diagram.

Description:

The use case diagram above reflects all three key user roles of the EskuuL platform (Student, Teacher, and Admin) shift and their interaction with the system. It graphically describes the big moves that all users make and the relationship between these moves with each other using the system capabilities.

Actors and Relationships:

- **Student**: The primary user. Students are able to log in to the system, obtain PDF summaries, take quizzes, see automatic feedback and see their academic progress. The purpose of these activities is to enhance learning and offer formal self-reflection.

- **Teacher:** The teachers will create and post educational resources, e.g., PDF summaries and quizzes. They also post materials to be examined, and the content uploaded should be in accord with the standards of curriculum.

- **Admin:** The administrator is a supervisor of the system. They operate user accounts, approve or deny the materials submitted, keep an eye on the system performance, and view content-related and student engagement analytics.

Connections and Flow:

- Login/Register: All actors begin by authenticating into the system.

- Access PDF Summaries and Take Quizzes: The students are involved directly with the course materials.

- Auto-Graded Feedback: Once a quiz is completed, results are displayed automatically, which allows the students to monitor the knowledge.

- Track Progress: Students are capable of seeing cumulative performance data on their quizzes.

- Create & Upload Content (PDFs and Quizzes): Teachers add new learning content.

- Send Materials to be reviewed: The uploaded materials are submitted to the admin to be moderated before they are published.

- Content Moderation & Approval: Admins will make sure that the materials are of quality and relevance and only then will they make it public.

- View Content Analytics: Admins are able to see system activity and engagement metrics.

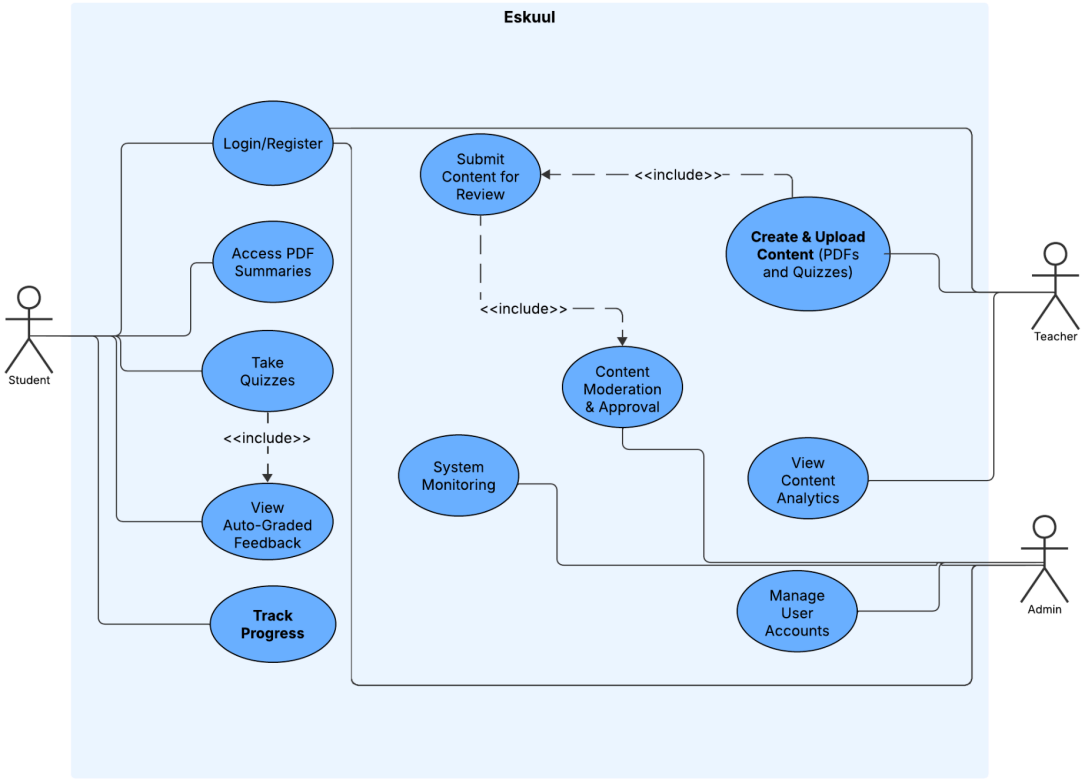
- User Accounts and System monitoring: Admins will ensure account security and system stability.

Input and Output:

- Input: User information, PDF uploads, quiz results, progress information.
- Outputs: Automated quiz marks, performance overviews, approved learning material and analytics on usage.

Purpose:

The diagram indicates how Teachers, students, and administrators collaborate in EskuuL ecosystem. The site will be created to guarantee quality delivery of content, organised learning and the ability to measure the progress.



B.3. Sequence Diagram Take and Grade Quiz.

Description:

The sequence diagram shows how the Student, the Web Interface, the Quiz Engine (Node.js API) and the Database (PostgreSQL) interact in a step-by-step fashion in case of a quiz attempt. It outlines the path of the data within the system since a student requests a quiz to the time the student gets his or her marked results and feedback.

Actors and Components:

- **Student:** The primary participant that starts and finishes the quiz.
- **Web Interface (Front-End):** The user interface made up of a web-based

platform where students engage with the platform. It processes any input action of the user and shows the response of the back end.

- **Quiz Engine (Node.js API):** The application logic that takes the requests, retrieves quiz information, grades answers, and sends results.
- **Database (PostgreSQL):** It contains all the information about the quiz such as questions, answers, correct answers, and results of the users.

Flow Explanation:

- **Request Quiz:**
  - The Student asks to commence a quiz by giving a quiz ID.
  - Quiz Engine receives this request at the Web Interface.
- **Fetch Quiz Questions:**
  - The Quiz Engine requests the Database with questions and answer options of the quiz.
  - Database replies with the desired information.
  - This quiz structure is sent by the API to the Web Interface which then shows the quiz to the student.
- **Submit Answers:**
  - Once the quiz is completed the Student then submits his/her answers.
  - The Web Interface transmits the results (attemptData) to the Quiz Engine where they are marked.
- **Grading and Feedback:**
  - The Quiz Engine fetches the accurate answers out of the Database.
  - It also provides a comparison of what the student responded to and the correct answers that had been stored and then generates a score and feedback.
  - The Web Interface in turn relays the results and feedback of the API.
  - The Web Interface shows the feedback and the score to the Student.

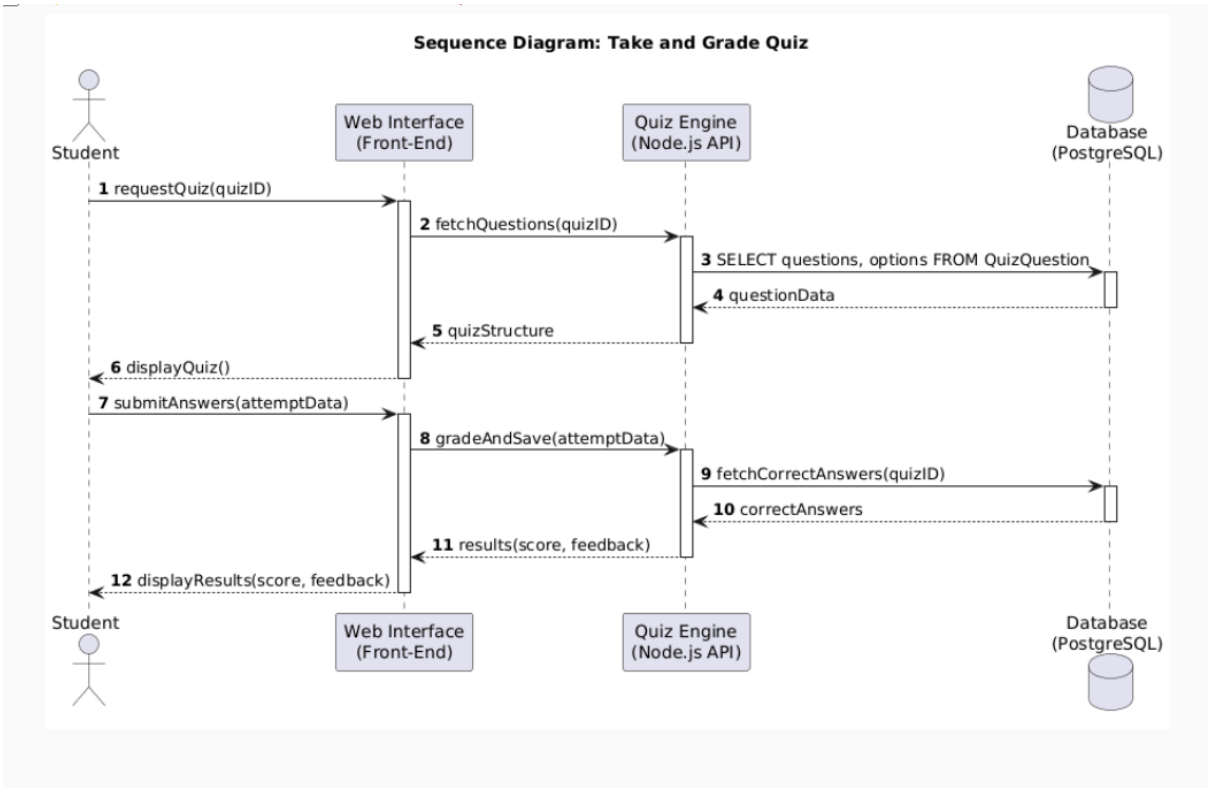
Input and Output:

- **Input:** quizID, answer data of the student.
- **Output:** evaluated results of the quiz, performance grade, and feedback.

Purpose:
The diagram shows the way in which the quiz subsystem facilitates real-time grading and feedback provision. It gives importance to automation and responsiveness and minimizes the workload of the teacher and delivers immediate feedback to the learners. The connection between the Node.js server and PostgreSQL provides an effective retrieval of the data, grading quality, and smooth student experience.

Relationships:

- The Web Interface serves as an intermediary between the Student and Quiz Engine.
- The Database is connected to the Quiz Engine which is the core logic layer.
- The Database has consistent quiz and result data, which is reliable and has integrity to data.



B.4. Activity Diagram - Review and Publishing Process of the Content.

Description:

The diagram below represents the end-to-end process of developing, revising, and publishing educational material (PDF summaries and quizzes) on the EskuuL platform. It outlines the process between three key parties such as the Teacher, the System, and the Admin describing the flow of content through the validation, review, and ultimate publication.

Actors and Swimlanes:

- **Teacher:** The one who is in charge of preparing educational materials, and passing them in order to be reviewed.

- **System:** Automated checks, validation, and storage as well as publication of approved contents.
- **Admin:** Checks the quality of content and makes sure that it complies with the national curriculum before approving the content.

Flow Explanation:

- **Creation and Submission of the Content:**
    - The procedure begins by the Teacher making PDF summaries or quizzes and posting them on the platform.
    -
    - As soon as the upload has been made, the teacher puts the material under examination.

- **System Validation:**
    - The System conducts first automated verification control to verify the technical requirements of the uploaded file including proper file format, metadata tagging and completeness.
    - In case of failure of the validation of file, the system throws out an error message and halts the operation to the teacher.
    - Provided that the content was valid, it goes to the Admin review phase.

- **Administrative Review:**
    - The Admin checks the content on quality assurance and verifies the curriculum.
    - In case the content is discovered unsuitable or inaccurate, the Teacher is informed of the reason why he or she has been rejected and the process terminates.
    - In case the content is approved, it is re-submitted to the System to be published.

- **Storage and Publishing:**
    - The **System** securely stores the approved content using cloud storage (e.g., AWS S3).

    - After successful storage, the content becomes publicly available for **Students** on the platform.

Inputs and Outputs:

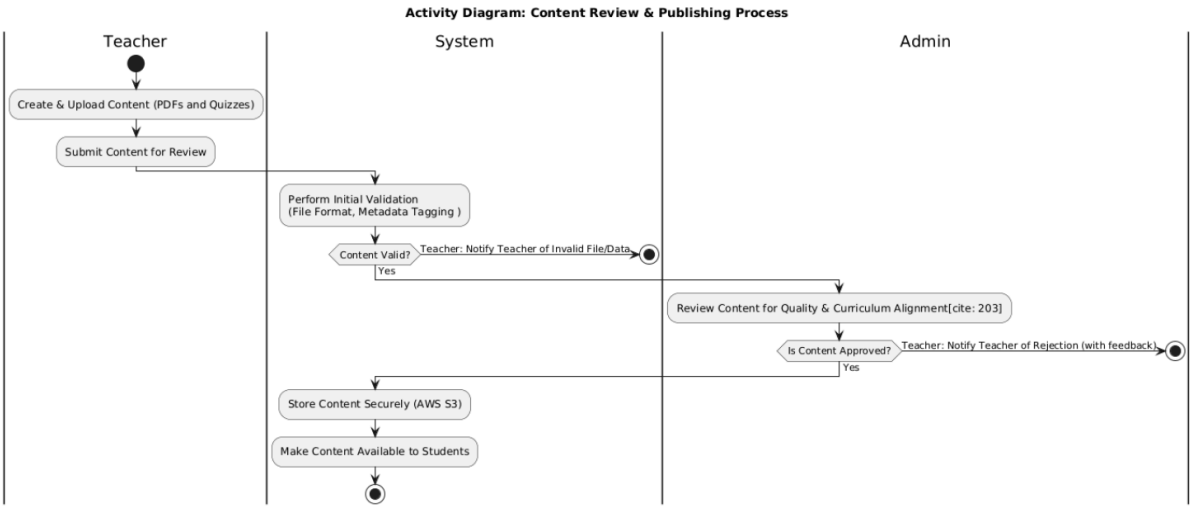
- Input: Uploaded content file, metadata information and review feedback.
- Output: Accepted learning content available to students.

Purpose:

As indicated in this diagram, EskuuL ensures the quality of the content and a high level of academic integrity by a well-organized process of moderation. It is facilitated by the fact that all resources that the students have access to are technically valid and accurate in the pedagogical sense.

Relationships and Activities:

- Teacher - System: Content upload and submission.
- System - Admin: Validate submissions to be reviewed by humans.
- Admin - Teacher: Improvement feedback in case of rejection.
- System: Manages the back end validation and the ultimate publishing to make it efficient and reliable.

Key Features Represented:

- The validation is automation to save the administrative time.
- Quality checks done by humans.
- Scalability with secure cloud storage.
- Open communication between the system, teachers and admins.